0580_w19_qp_22_Q22

22 $\mathbf{A} = \begin{pmatrix} 3 & 2 \\ -5 & 0 \end{pmatrix}$ $\mathbf{B} = \begin{pmatrix} -2 & 5 \\ 4 & 1 \end{pmatrix}$ $\mathbf{C} = \begin{pmatrix} -1 & k \end{pmatrix}$

(a) Find **AB**.

$\begin{pmatrix} \quad & \quad \\ \quad & \quad \end{pmatrix}$ [2]

(b) $\mathbf{CA} = \begin{pmatrix} -13 & -2 \end{pmatrix}$

Find the value of k.

$k =$ ..... [2]

(c) Find $\mathbf{A}^{-1}$.

$\begin{pmatrix} \quad & \quad \\ \quad & \quad \end{pmatrix}$ [2]